Numerics II
Freie Universität Berlin
Winter semester 2025/2026
Dr. Robert Gruhlke, André-Alexander Zepernick

Exercise Sheet 5

Deadline: Friday, November 21st, 2025 at 10 am

Exercise 1 (3-step Adams-Bashforth method; 4 points)
Derive the 3-step Adams-Bashforth method from the formulas in the lecture notes.

Programming Exercise 1 (Implementation of embedded RK schemes; $4 + 4 = 8$ points)
In this exercise we extend the numerical integration of the van der Pol system, which was already investigated on exercise sheet 4, i.e.,

$$\begin{aligned} y_1'(t) &= y_2(t), \\ y_2'(t) &= 10\,(1 - y_1(t)^2)\,y_2(t) - y_1(t), \qquad t \in (0, 20], \\ y_1(0) &= 0, \qquad y_2(0) = 1, \end{aligned}$$

by implementing an *adaptive step–size control* based on an embedded Runge–Kutta scheme.
The adaptive controller follows the procedure:

Algorithm 1 Adaptive Step–Size Control

Require: current step size h, current value Y_k at time t_k, increment functions Φ and $\widetilde{\Phi}$ of order q and q+1, and an error tolerance $\varepsilon > 0$.
Ensure: next value Y_{k+1} and next step size estimate h_{next}.

1: Compute the two trial solutions

$$Y = Y_k + h\,\Phi(t_k, Y_k, h), \qquad \widetilde{Y} = Y_k + h\,\widetilde{\Phi}(t_k, Y_k, h).$$

2: Estimate the scaling factor

$$s = \left(\frac{h\,\varepsilon}{\|Y - \widetilde{Y}\|_\infty} \right)^{1/q}.$$

3: **if** $s \geq 1$ **then**
4: Accept the step: $Y_{k+1} \leftarrow \widetilde{Y}$, $t_{k+1} \leftarrow t_k + h$.
5: Update the step size estimate $h_{\text{next}} \leftarrow \min\{2, s\}\,h$.
6: **else**
7: Reject the step and repeat with $h \leftarrow \max\{1/2, s\}\,h$.
8: **end if**

a) Implement this algorithm in Python using the *embedded* RK4(3) scheme, which you derived on exercise sheet 4. Your function should have the interface

`solve_adaptive_rk43_procedure(f, y0, I, h0, epsilon, q)`,

where `f` defines the right–hand side of the ODE, `y0` is the initial value, and $I = [a, b]$ is the integration interval.

b) Apply your implementation to the van der Pol system on $I = [0, 20]$ with parameters

$$h_0 = 10^{-4}, \qquad \varepsilon = 10^{-4}.$$

- Plot the components $y_1(t)$ and $y_2(t)$ versus t.
- Compare the performance and accuracy of your adaptive scheme with a fixed–step classical RK4 method using step size $h = 0.0625$.
- Compute and plot a reference solution (e.g. classical RK4 with $h_{\text{ref}} = 2^{-14}$) and determine the maximal global error

$$\max_k \|Y_k - y_{\text{ref}}(t_k)\|_\infty$$

for both the adaptive and the fixed–step schemes. Report the total number of function evaluations and the observed errors.

Programming Exercise 2 (Adams-Bashforth method; 1 + 2 + 1 = 4 points)
Consider the initial value problem from exercise sheet 2, i.e.,

$$y'(t) = -200\, t\, y(t)^2, \qquad t \in [-1, 0], \qquad y(-1) = \frac{1}{101},$$

whose exact solution is given by

$$y(t) = \frac{1}{100t^2 + 1}.$$

a) Recall that the Adams–Bashforth method of order $m = 2$ for a constant step size h reads

$$Y_{k+1} = Y_k + h\left(\frac{3}{2} f(t_k, Y_k) - \frac{1}{2} f(t_{k-1}, Y_{k-1})\right), \qquad k = 1, \ldots, N-1.$$

Since this is a two-step method, use the explicit Euler method to compute Y_1 from the initial condition:

$$Y_1 = Y_0 + h\, f(t_0, Y_0), \qquad Y_0 = y(-1) = \frac{1}{101}.$$

b) Implement this method in `Python` for the given problem on the interval $[-1, 0]$. Use a uniform grid with step size $h = \frac{1}{N}$ and grid points

$$t_k = -1 + kh, \qquad 0 \le k \le N,$$

for $N \in \{25, 50, 100, 200, 400, 800, 1600\}$.

c) For each N, compute the numerical solution $\{Y_k\}$ and the exact solution $\{y(t_k)\}$. Determine the maximum error

$$E_N = \max_{0 \le k \le N} |Y_k - y(t_k)|$$

and verify numerically that the Adams–Bashforth method of order two exhibits second-order convergence.

Rules for submission & passing criterion:
Exercises must be completed in **groups of three** and **submitted electronically via the Whiteboard system** under "**Assignments**." Late submissions will not be corrected. Please **state the names of all group members** on the submission. The programming language of this course is **Python**. The exercises will be discussed in the tutorial the following week. To pass the tutorial component of the module, you must achieve **at least 50% of the homework points**.